

Project Proposal: ExpenseShare (Splitwise Clone)

Date: October 12, 2025
Platform: iOS Native (iPhone)
Initial Scope: Minimum Viable Product (MVP)

1. Executive Summary

This proposal outlines the development of **ExpenseShare**, a robust and intuitive native iOS application designed to eliminate the awkwardness and stress associated with sharing and settling expenses among friends, family, and groups. ExpenseShare will provide a clear, real-time ledger for tracking debts, logging payments, and simplifying complex bill-splitting scenarios, focusing on a clean, modern iOS-specific user experience using SwiftUI. The core value proposition is to offer complete financial transparency and automatically calculate the simplest way to settle debts, ensuring all parties know "who owes who" at any given time.

2. Project Goals and Objectives

- **Primary Goal:** Successfully launch a functional Minimum Viable Product (MVP) on the Apple App Store within 18-20 weeks.
- **User Experience:** Deliver a fast, stable, and highly intuitive user interface (UI/UX) that adheres strictly to Apple's Human Interface Guidelines.
- **Data Integrity:** Ensure 100% accuracy in all expense calculations, balances, and debt simplification algorithms.
- **Scalability:** Implement a scalable cloud-based backend (Firestore) to support real-time data sync across multiple users and large groups.

3. Technology Stack

Component	Technology	Rationale
Frontend/UI	Swift (Native iOS) / SwiftUI	Ensures high performance, modern design language, and deep integration with the iOS ecosystem.
Backend/Database	Firebase Firestore	Provides real-time, scalable data synchronization and NoSQL storage suitable for group-based transactions.
Authentication	Firebase Auth	Secure user sign-up, login (Email/Password), and

		anonymous access capability.
Deployment	Apple App Store	Final deployment target.

4. Core Features and User Functionalities (MVP Scope)

The following list details the mandatory features required for the MVP, categorized by user action.

A. User Management and Setup

Feature	User Should Be Able To...
Account Creation	Register and log in securely using email/password.
Friend Management	Add and invite friends using email or unique username/ID.
User Profile	View and edit their personal profile, including a custom avatar and name.
Real-Time Balances	View a main dashboard showing the aggregated total amount they owe and the total amount they are owed across all groups and private expenses.

B. Group and Expense Tracking

Feature	User Should Be Able To...
Group Creation	Create distinct groups (e.g., "Paris Trip 2025," "Apartment Rent," "Dinner") with custom names and covers, inviting friends to join.
Add Expense	Log a new expense within a group, including the total amount, description, date, and category.
Designate Payer	Specify which member of the group paid for the expense (defaulting to the user logging the expense).
Attach Proof	Add a photo of the receipt or invoice for transparency (stored in Firebase Storage).
Expense History	View a chronological list of all expenses and activities within any selected group.

C. Advanced Splitting Logic

Feature	User Should Be Able To...
Equal Split	Automatically divide the expense equally

	among all selected group members.
Unequal Split (Custom Amounts)	Input custom, exact dollar amounts that each specific member owes.
Unequal Split (Percentages/Shares)	Split the bill based on custom percentages or units/shares, allowing for complex divisions (e.g., Jane owes \$20, Mike owes 40%, and the remaining 60% is split equally between the others).
Exclude Members	Log an expense where a group member was present but should not be charged (e.g., a non-drinker at a bar bill).

D. Settlement and Debt Simplification

Feature	User Should Be Able To...
Settle Up	Record a payment from one member to another (e.g., "Sarah paid \$50 cash to Mike").
Debt Simplification	The app must run an algorithm to show the simplest path to settle all debts (e.g., instead of A paying B, and B paying C, simplify to A paying C).
Private Debts	Log a simple, non-group debt between two individuals (an IOU).
Activity Feed	View a global activity log of all new expenses, edits, and settlements across all groups and friendships.

5. High-Level Project Timeline

The project will be executed in five phases, optimized for iterative development.

Phase	Description	Estimated Duration
Phase 1: Discovery & Planning	Finalizing wireframes, creating UI mockups, and defining the complete technical architecture and database schema.	3 Weeks
Phase 2: Core Infrastructure	Setting up Firebase Auth, Firestore data models, user registration, and friend/group creation.	4 Weeks
Phase 3: Expense Logic & UI	Developing the "Add Expense" screen, implementing all	6 Weeks

	splitting algorithms, and building the real-time group ledger view.	
Phase 4: Settlement & QA	Implementing the debt simplification and settlement recording functionality. Dedicated bug-fixing, performance tuning, and internal QA testing.	4 Weeks
Phase 5: Beta & Deployment	Conducting external beta testing (TestFlight), creating App Store assets (screenshots, description), and submitting the final build for review.	3 Weeks

Total Estimated Duration (MVP Launch): 20 Weeks

6. Next Steps

Upon approval of this proposal, the next steps will involve a deep-dive design session to finalize the brand style guide and create high-fidelity screens based on the approved feature set.